

**PLC Wiring Drawings:
Practical Interpretation and Implementation
(PWD-PII)**

(Lab Work One)

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INTRODUCTION

Lab Work One Overview – Day 1

- Explain Automations and AIs revolution. And the meaning of PLC.
- P – Programmable.... Programing languages e.g. LAD or LD, FBD or FD
Programing Software e.g. TIA Portals V13, V14, V18
- L – Logic.... Basics: True, False, On, Off, 0, 1
Boolean Algebra $y = 1$ and $\bar{y} = 0$
- C – Controllers PLC's CPU, Sinamics (VFD Drive), and Human Machine Interface (HMI)

PLCs BRIEF ARCHITECTURE

Day 1 cont.

- Figure 1.0 shows the basic architecture for Programmable Logic Controllers (PLC) .

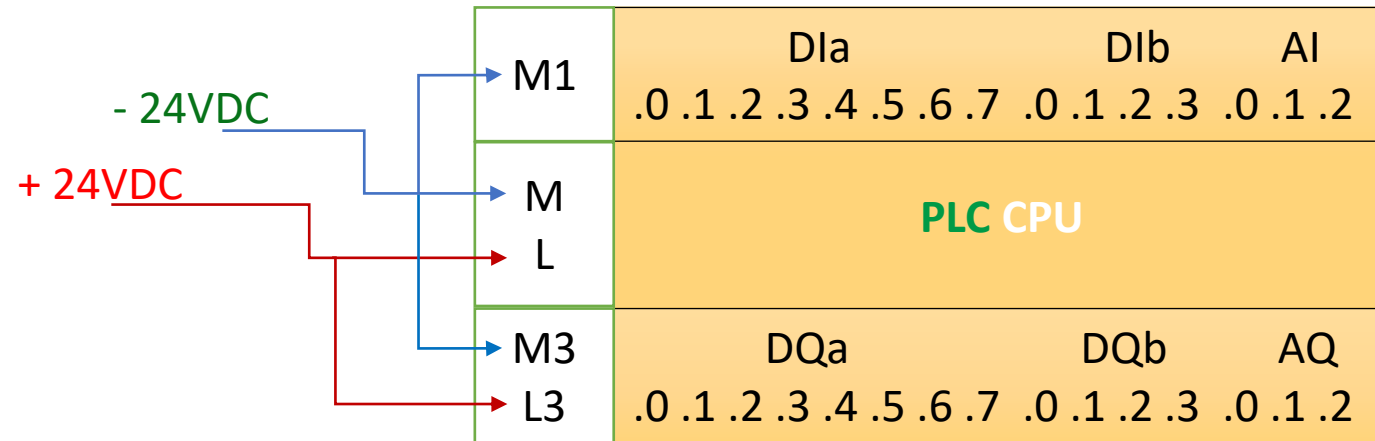


Figure 1.0: Basic architecture for Programmable Logic Controllers (PLC)

- Demonstrate 1.0: After Explanation, Show students these sections with a Step 7 1200 or 1400 PLC*

UEDS

Day 1 Cont.

Control Circuit and Power Circuit Symbols

Overloads

Thermal



Contacts

Normally Closed



Normally Open

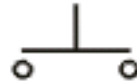


Push Buttons

Normally Closed



Normally Open



Switches

Timing Closed NC



Timing Closed NO



Timing Open NC



Timing Open NO



Switches

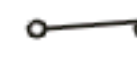
Flow NC



Flow NO



Normally Closed



Normally Open



Limit NC



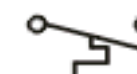
Limit NO



Temperature NC



Temperature NO



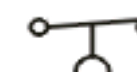
Pressure NC



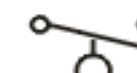
Pressure NO



Liquid NC



Liquid NO



Fuse

Generic



Coil

Generic



Circuit Breaker

Generic



Motor

Three Phase



Indicator Light

Generic



Transformer



UEDS

Day 1 Cont.

- *Demonstrate 2.0: After Explanation, Use Multi Meter to demonstrate the NO and NC parts.*
- *Demonstrate 2.1: wire a manual control with an indicator light, Push buttons and logo.*
- *Demonstrate 2.2: Do 2.0 again.*
- **Report 1:** Write a 2-3 page report covering Day 1 and 2.0, 2.1, and 2.2. Add your observations

REFERENCES

[1] Mr. Daniel Kwagyir “Course Outline: Electrical Drawings”, Electrical & Electronic Engineering Department, KNST

[2]

